

WEEK 3 ASSIGNMENT

PART A:

1. Swapping of Two Numbers

Write a C program to swap two numbers using a temporary variable.

o Input: Two integers.

o Output: The values after swapping.

Example:

Input: a = 10, b = 20

Output: a = 20, b = 10

```
1  /* Swaping Real Numbers*/
2  #include<stdio.h>
3  int main()
4  {
5      int a=10,b=20,swap;
6      swap=a;
7      a=b;
8      b=swap;
9      printf("Values of a,b=%d,%d",a,b);
10     return 0;
11 }
```

2. Find Sum, Difference, Product, Quotient, and Remainder

Write a C program to input two integers and compute:

o Sum

o Difference

o Product

o Quotient

o Remainder

Example:

Input: a = 15, b = 6

Output: Sum = 21, Difference = 9, Product = 90, Quotient = 2,

Remainder = 3

```
1  /*Algebraic expressions */
2  #include<stdio.h>
3  int main()
4  {
5      int a=15,b=7,c,d,e,f,g;
6      printf("Sum=%d\n", (a+b));
7      printf("Difference=%d\n", (a-b));
8      printf("Product=%d\n", (a*b));
9      printf("Quotient=%d\n", (a/b));
10     printf("Remainder=%d", (a%b));
11     return 0;
12 }
```

3. Calculate Body Mass Index (BMI)

Write a C program to calculate the BMI of a person.

Formula:

$BMI = \text{weight (kg)} / (\text{height (m)})^2$

Input: Weight in kilograms, Height in meters.

Output: BMI value and category (Underweight, Normal, Overweight, Obese).

Example:

Input: Weight = 70, Height = 1.75

Output: BMI = 22.86 → Normal

```

1  /**Body Mass Index */
2  #include<stdio.h>
3  int main()
4  {
5      float h,w,BMI;
6      printf("Height of person=");
7      scanf("%f",&h);
8      printf("Weight of person=");
9      scanf("%f",&w);
10     BMI=w/(h*h);
11     if(BMI<18.5)
12     {
13         printf("Underweight");
14     }
15     else if(BMI>=18.5 && BMI<25)
16     {
17         printf("Normal");
18     }
19     else if(BMI>=25 && BMI<30)
20     { printf("Overweight");
21     }
22     else
23     { printf("Obese");
24     }
25     return 0;
26 }

```

PART B:

a)

```
printf("Enter a number: ");
```

```
scanf("%d",&number);
```

```

1  #include<stdio.h>
2  int main()
3  {
4      int number;
5      printf("Enter a number:");
6      scanf("%d",&number);
7      return 0;
8  }

```

b)

printf("Value of x is: %d", x, y);

```
1  /*Correction of Sentences*/  
2  #include<stdio.h>  
3  int main()  
4  {  
5      int x=10,y=15;  
6      printf("Value of x is=%d",x);  
7      return 0;  
8  }
```

c)

scanf("%f", &a, &b);

```
1  /*Correction of Sentences*/  
2  #include<stdio.h>  
3  int main()  
4  {  
5      float a,b;  
6      printf("Values of a,b=");  
7      scanf("%f %f",&a,&b);  
8  
9      return 0;  
10 }
```

d)

printf("Sum of %d and %d is %d", a, b);

```

/*Correction of Sentences*/
#include<stdio.h>
int main()
{
    int a,b;
    printf("value of a:");
    scanf("%d",&a);
    printf("value of b:");
    scanf("%d",&b);
    printf("the sum of %d and %d is %d",a,b,a+b);
    return 0;
}

```

e)

```
scanf("%d %d", a, b);
```

```
printf("The average is %f", (a+b)/2);
```

```

/*Correction of Sentences*/
#include<stdio.h>
int main()
{
    float a,b;
    printf("Enter the first number:");
    scanf("%f",&a);
    printf("Enter the second number:");
    scanf("%f",&b);
    printf("the average is %f",((a+b)/2));
    return 0;
}

```

f)

```
scanf("%C", &str);
```

```

1  /*Correction of Sentences*/
2  #include<stdio.h>
3  int main()
4  {
5      char ch;
6      printf("character=");
7      scanf("%c",&ch);
8      printf("character is %c",ch);
9      return 0;
10 }

```

g)

printf("The value is: %d", &x);

```

1  /*Correction of Sentences*/
2  #include<stdio.h>
3  int main()
4  {
5      int x;
6      printf("value of x=");
7      scanf("%d",&x);
8      printf("The value is:%d",x);
9      return 0;
10 }
11

```

-THE END-